

# SuwaMed: A Comprehensive Telehealth Platform

*Bridging the Rural-Urban Healthcare Divide in Sri Lanka*

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## Abstract

Access to quality healthcare in Sri Lanka remains sharply uneven between urban and rural communities. Patients in rural districts routinely travel two to four hours and incur costs exceeding LKR 2,000 for a single consultation, while qualified doctors remain concentrated in major urban centres. Existing international telemedicine platforms such as oDoc and Practo address only part of the problem; they assume high-speed connectivity, lack support for Sinhala and Tamil, and do not integrate with local payment infrastructure or the Sri Lanka Medical Council (SLMC) verification process.

This project presents **SuwaMed**, a full-stack telehealth platform purpose-built for the Sri Lankan healthcare ecosystem. The system comprises a cross-platform mobile application developed with React Native and Expo and a Node.js/TypeScript backend built on Express and MongoDB. Three role-based interfaces are provided for patients, doctors, and administrators. Core features include an AI-powered symptom checker (OpenAI) operating in English, Sinhala, and Tamil; real-time video and chat consultations powered by Agora and Socket.io; appointment scheduling with automated reminders via Firebase Cloud Messaging; encrypted health record storage through Cloudinary; digital prescription generation; and an admin-driven SLMC verification workflow. Local payments are processed in LKR through PayHere, with subscription tiers and doctor-earnings withdrawal supported.

Security is enforced through JWT access tokens with refresh-token rotation, bcrypt password hashing, OTP-based phone and email verification, role-based access control, and rate-limited authentication endpoints. The architecture is optimised for low-bandwidth (2G/3G) environments common in rural Sri Lanka. By combining trilingual support, locally relevant payment integration, verified doctor onboarding, and AI-assisted triage in a single platform, SuwaMed seeks to reduce diagnostic delay, lower the financial and logistical cost of consultation, and extend specialist expertise to underserved populations.

**Keywords:** Telemedicine, Telehealth, Sri Lanka, AI Symptom Checker, Mobile Health (mHealth), React Native, Node.js, MongoDB, Multilingual Healthcare, Rural Healthcare Access.